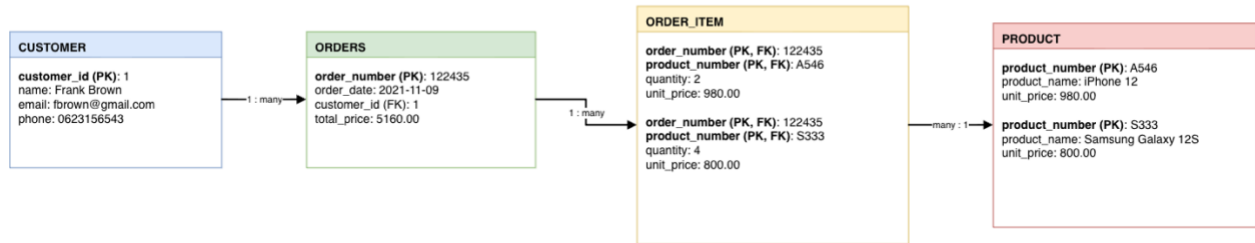


1. Relational Database Design

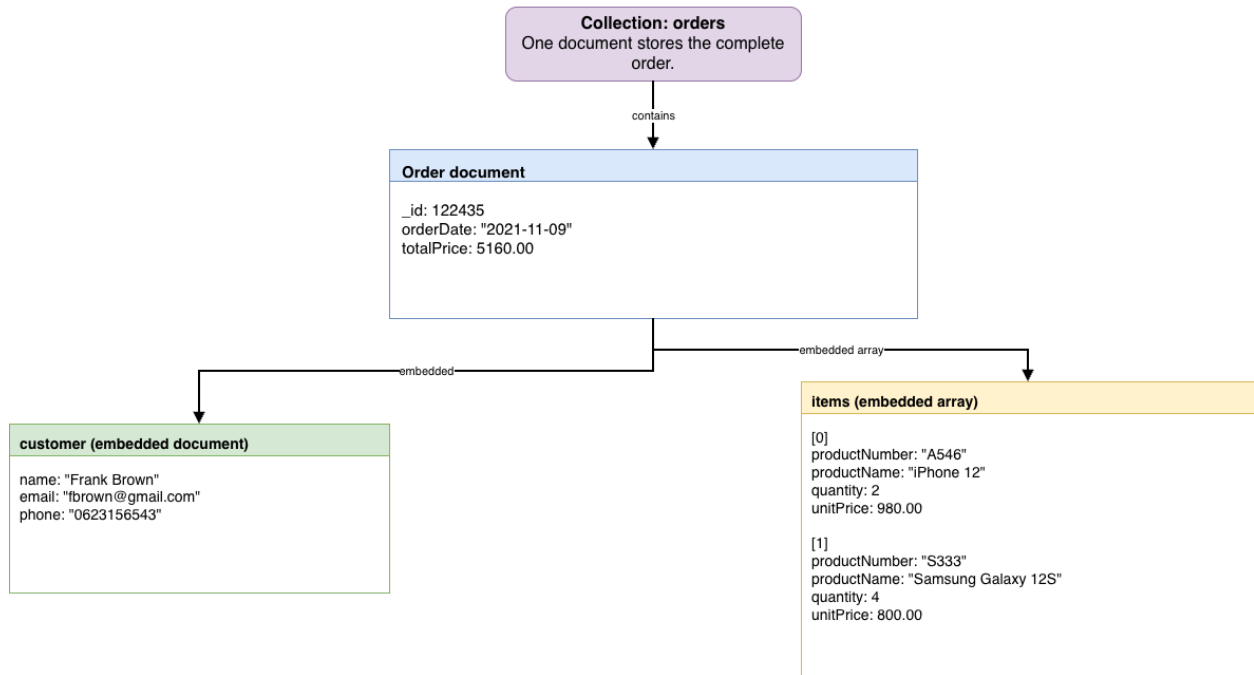
Relational Database Design — Order 122435



Note: unit_price is stored in ORDER_ITEM to preserve the purchase-time price.

2. MongoDB Database Design

MongoDB Design — storeDB.orders Collection



3. Cassandra Database Design

Cassandra Design — Query: Retrieve an Order and Its Items by Order Number

Keyspace: store
Table: orders_by_order_number

orders_by_order_number
order_number (Partition Key): int product_number (Clustering Key): text order_date: date customer_name: text customer_email: text customer_phone: text total_price: decimal product_name: text quantity: int unit_price: decimal
PRIMARY KEY (order_number, product_number)

Query supported
SELECT * FROM store.orders_by_order_number
WHERE order_number = 122435;

Stored rows

Row 1

order_number: 122435
product_number: A546
order_date: 2021-11-09
customer_name: Frank Brown
customer_email: fbrown@gmail.com
customer_phone: 0623156543
total_price: 5160.00
product_name: iPhone 12
quantity: 2
unit_price: 980.00

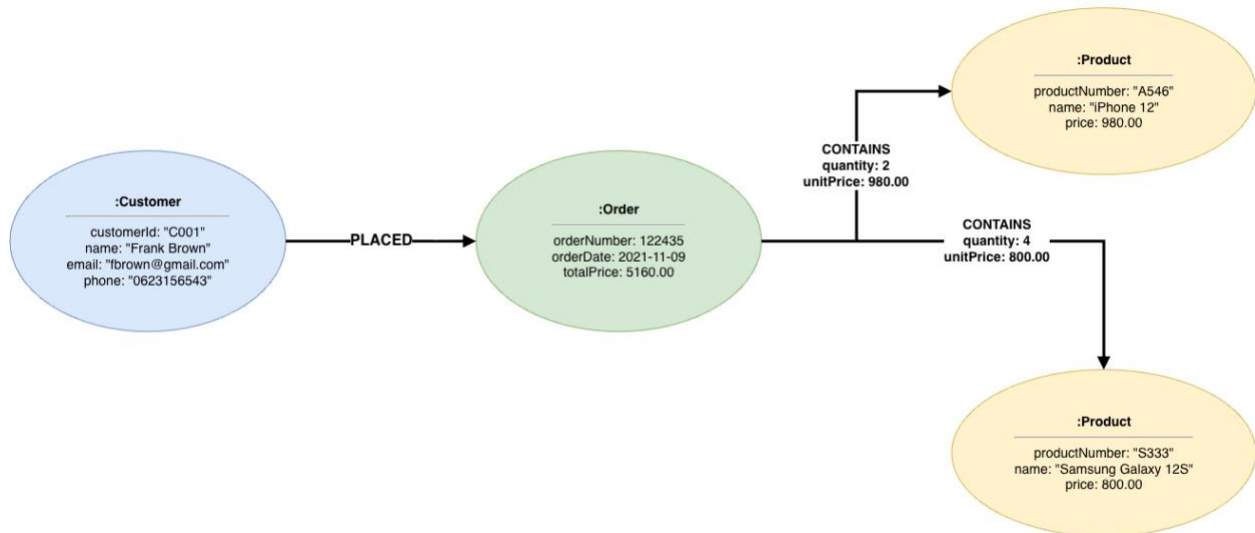
Row 2

order_number: 122435
product_number: S333
order_date: 2021-11-09
customer_name: Frank Brown
customer_email: fbrown@gmail.com
customer_phone: 0623156543
total_price: 5160.00
product_name: Samsung Galaxy 12S
quantity: 4
unit_price: 800.00

Order and customer values repeat for each item row by design (query-based denormalization).

4. Neo4j Database Design

Neo4j Property Graph Design — Order 122435



Line-item values quantity and unitPrice are relationship properties because they describe a specific Order-Product connection.